

Report for the Problem Set 3

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Introduction

The goal of this project is to extract information about how stock returns are distributed, using option prices instead of historical data. We follow the method of Bakshi, Kapadia, and Madan (2003).

Task 1

The first step was to compute three intermediate quantities (V, W, and X) directly from the option prices. These are obtained by weighting each option price by a factor that depends on how far the strike is from the current stock price, and then summing up these weighted prices across all available strikes. Out-of-the-money put options (which pay off in a crash) and out-of-the-money call options (which pay off in a rally) contribute separately, each with their own weighting formula.

The key difference between V, W, and X is that each one captures a higher-order feature of the distribution: V is mostly about variance, W is skewness, and X is sensitive to extreme tail events (kurtosis).

Once V, W, and X were computed for every stock-day combination, we combined them with the 30 day risk-free rate to produce five implied moment quantities: the expected return, the variance, the annualized volatility, the skewness, and the kurtosis. This was done for both the SPX index and all 500 S&P 500 parties (9471 rows in total).

Results:

[5]:	Symbol	loctimestamp	daystomaturity	V_t_tau	W_t_tau	X_t_tau
0	A	2023-02-01	30	0.008646	-0.000621	0.000450
1	A	2023-02-02	30	0.009058	-0.000705	0.000463
2	A	2023-02-03	30	0.008755	-0.000751	0.000470
3	A	2023-02-06	30	0.009360	-0.000782	0.000513
4	A	2023-02-07	30	0.009339	-0.001164	0.000642

[4]:	Symbol	loctimestamp	daystomaturity	V_t_tau	W_t_tau	X_t_tau
0	^SPX	2023-02-01	30	0.002881	-0.000274	0.000111
1	^SPX	2023-02-02	30	0.003308	-0.000348	0.000150
2	^SPX	2023-02-03	30	0.002807	-0.000274	0.000108
3	^SPX	2023-02-06	30	0.003302	-0.000335	0.000144
4	^SPX	2023-02-07	30	0.002921	-0.000285	0.000119

Overall, the BKM integrals reveal that the individual stock exhibits higher implied variance, stronger asymmetry and larger tail-risk measures than the S&P500 index. This is consistent with financial theory, as individual stocks are exposed to both market-wide and firm-specific risks, whereas the index benefits from diversification.

[7]:	Symbol	loctimestamp	daystomaturity	bakshi_mu	bakshi_implVol	bakshi_implVol_annual	bakshi_skew	bakshi_kurt
0	*SPX	2023-02-01	30	0.002449	0.002886	0.187392	-1.913569	13.707432
1	*SPX	2023-02-02	30	0.002274	0.003315	0.200836	-1.947115	14.026021
2	*SPX	2023-02-03	30	0.002444	0.002812	0.184953	-1.983594	14.000475
3	*SPX	2023-02-06	30	0.002251	0.003310	0.200675	-1.883174	13.448459
4	*SPX	2023-02-07	30	0.002434	0.002927	0.188704	-1.939267	14.272960
5	*SPX	2023-02-08	30	0.002297	0.003261	0.199179	-1.984725	14.280680
6	*SPX	2023-02-09	30	0.002096	0.003718	0.212689	-1.964249	13.942713
7	*SPX	2023-02-10	30	0.002207	0.003433	0.204380	-1.973603	14.307200
8	*SPX	2023-02-13	30	0.002036	0.003885	0.217420	-1.886176	13.740772
9	*SPX	2023-02-14	30	0.002452	0.002883	0.187293	-1.990065	14.390981
10	*SPX	2023-02-15	30	0.002533	0.002745	0.182742	-1.980020	14.324789
11	*SPX	2023-02-16	30	0.002362	0.003235	0.198382	-2.268496	16.074111
12	*SPX	2023-02-17	30	0.002417	0.002971	0.190128	-2.124355	14.919006
13	*SPX	2023-02-21	30	0.001848	0.004200	0.226059	-2.129845	15.502198
14	*SPX	2023-02-22	30	0.001900	0.004051	0.222504	-2.185593	15.707954
15	*SPX	2023-02-23	30	0.002277	0.003420	0.203973	-1.968802	14.030799
16	*SPX	2023-02-24	30	0.002189	0.003633	0.207321	-1.967720	14.889930
17	*SPX	2023-02-27	30	0.002257	0.003954	0.208233	-1.915855	14.860501
18	*SPX	2023-02-28	30	0.002342	0.003363	0.202281	-1.835363	14.180477

[8]:	Symbol	loctimestamp	daystomaturity	bakshi_mu	bakshi_implVar	bakshi_implVol_annual	bakshi_skew	bakshi_kurt
0	A	2023-02-01	30	-0.000401	0.008679	0.324955	-0.757728	5.984801
1	A	2023-02-02	30	-0.000566	0.009093	0.332611	-0.798771	5.606520
2	A	2023-02-03	30	-0.000476	0.008788	0.326990	-0.899281	6.096480
3	A	2023-02-06	30	-0.000730	0.009396	0.338109	-0.839692	5.810492
4	A	2023-02-07	30	-0.000662	0.009375	0.337730	-1.266905	7.296843

The results show clear differences between the SPX index and the individual stock. The annualized implied volatility of the SPX ranges from approximately 18% to 23%, while stock A exhibits considerably higher values of around 32% to 34%. This suggests that investors expect much greater uncertainty for the individual stock than for the diversified market index.

Both assets exhibit negative implied skewness, indicating that option markets assign a higher probability to large negative returns than to large positive returns. The SPX skewness is particularly negative, ranging from about -1.9 to -2.3, meanwhile stock A ranges between roughly -0.8 and -1.3. This suggests that investors are especially concerned about market-wide downturns. Furthermore, the implied kurtosis values are substantially above the normal-distribution benchmark of three. For stock A, kurtosis lies mostly between 5 and 7, while the SPX reaches values between 13 and 16. These high values indicate fat tails and imply that extreme outcomes are considered much more likely than under a normal distribution. Overall, the results suggest that option markets price significant downside and tail risk, while diversification reduces the overall volatility of the market index compared to individual stocks.

Task 2

We split each stock's implied skewness into two parts: a systematic part that comes from the stock's exposure to the overall market, and an idiosyncratic part that is specific to the individual company. The logic comes from the CAPM, which says that a stock's return is a combination of a market-driven component and a firm-specific component. Because these two components are independent of each other, their contributions to skewness can be separated. The market-driven contribution is proportional to the stock's beta cubed, meaning even a moderately high beta results in a small systematic skewness contribution, since cubing a number less than one makes it much smaller.

Results:

[9]:	Symbol	loctimestamp	daystomaturity	bakshi_skew	bakshi_skew_sys	bakshi_skew_eps
0	A	2023-02-01	30	-0.757728	-0.008648	-0.749079
1	A	2023-02-02	30	-0.798771	-0.001794	-0.796977
2	A	2023-02-03	30	-0.899281	-0.019565	-0.879716
3	A	2023-02-06	30	-0.839692	-0.025727	-0.813966
4	A	2023-02-07	30	-1.266905	-0.090274	-1.176632
5	A	2023-02-08	30	-1.316317	-0.009003	-1.307315
6	A	2023-02-09	30	-1.047887	-0.001893	-1.045993
7	A	2023-02-10	30	-0.558790	-0.000101	-0.558689
8	A	2023-02-13	30	-1.919333	-0.022944	-1.896389
9	A	2023-02-14	30	-1.237708	-0.247283	-0.990425

The skewness decomposition shows that most of the negative skewness of stock A comes from idiosyncratic risk rather than systematic market risk. The total implied skewness ranges from about -0.6 to -1.9 during the sample period. However, the systematic component is relatively small, while the idiosyncratic component explains almost all of the total skewness. For example, on 2023-02-01 the total skewness is -0.758, but only -0.009 comes from systematic risk,

meanwhile -0.749 comes from idiosyncratic risk. Similar patterns can be observed on most other dates. This suggests that investors mainly associate the downside risk of stock A with firm-specific factors rather than overall market conditions.

Task 3

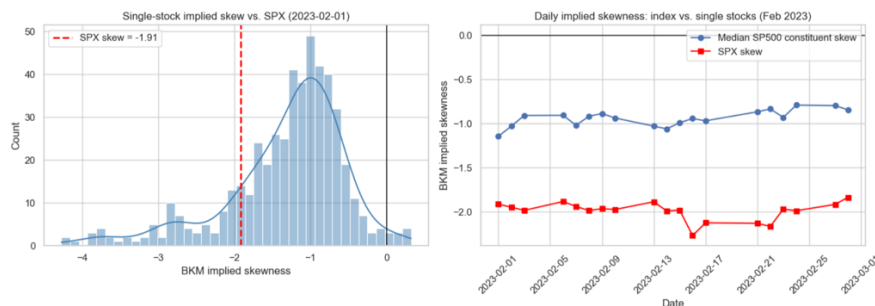
Results:

(500, 3)

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[11]:
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	Symbol	Percentage_Skew_less_than_0	Percentage_Skew_more_than_SPX
0	A	100.0	89.473684
1	AAL	100.0	100.000000
2	AAP	100.0	100.000000
3	AAPL	100.0	100.000000
4	ABBV	100.0	100.000000

The results show that the implied skewness of many individual stocks is negative throughout the entire sample period. For example, stocks such as A, AAL, AAP, AAPL, and ABBV have negative skewness on 100% of the observed days. This indicates that option markets consistently assign a higher probability to large negative returns than to large positive returns. At the same time, these stocks usually have higher skewness values than the SPX index. For example, stock A has a skewness greater than the SPX on approximately 89% of the trading days. This suggests that although individual stocks exhibit downside risk, the market index generally has even more negative skewness, reflecting stronger concerns about broad market downturns and crash risk.



The figures show that the SPX index is more negatively skewed than individual stocks. In the histogram, the SPX skewness (-1.91) lies to the left of most stock skewness values. The time-series plot confirms that this relationship is stable over the whole sample period. Therefore, the graphical evidence supports the numerical results and the main finding of BKM.